

親水性壓克力人工水晶體在無併發症 nDSAEK 後產生前表面混濁：病例報告

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Anterior Opacification of a Hydrophilic Acrylic Intraocular Lens Following Uncomplicated Non-Descemet’ s Stripping Automated Endothelial Keratoplasty: A Case Report

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Introduction

Intraocular lens (IOL) opacification is a rare but visually debilitating late complication following intraocular surgery. We report a distinctive case of localized anterior surface opacification of a hydrophilic acrylic IOL following uncomplicated non-Descemet’ s stripping automated endothelial keratoplasty (n-DSAEK) with intracameral air tamponade.

Case Presentation

Patient Profile: 75-year-old male with a history of heart failure with preserved ejection fraction (on Bisoprolol) and benign prostatic hyperplasia (on Tamsulosin).

Ocular History: Pseudophakic bullous keratopathy in the right eye (OD) following cataract surgery with a hydrophilic acrylic IOL (Rayner) performed elsewhere in July 2024.

Preoperative Baseline (February 2025):
Best-Corrected Visual Acuity (BCVA): 0.1 OD, 0.4 OS.
Central Corneal Thickness (CCT): 877 μm OD (marked corneal edema), 518 μm OS (clear).

Surgical Intervention (March 2025): Uncomplicated n-DSAEK with intracameral air injection for graft tamponade OD.

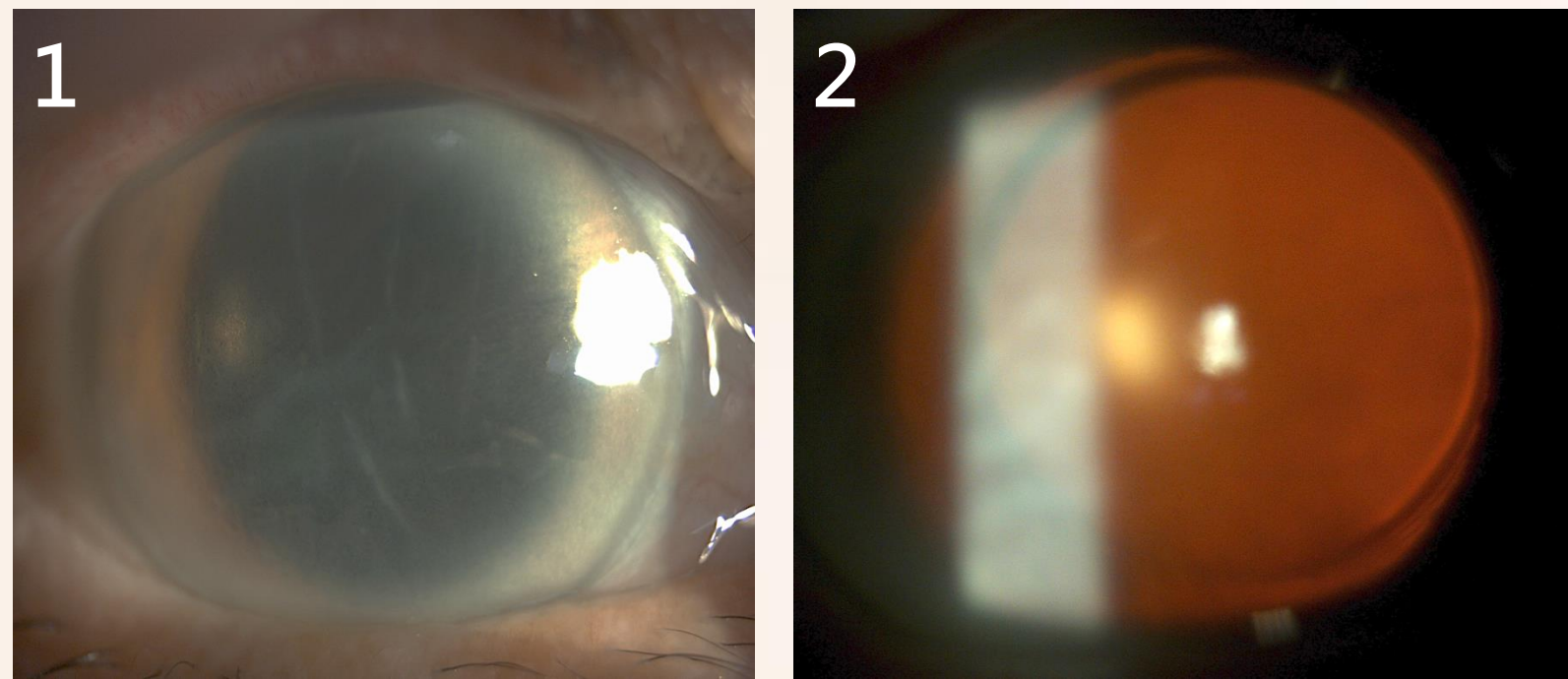
Postoperative Recovery (July 2025):
Cornea cleared successfully; BCVA OD improved to 0.4 with CCT 518 μm.
Endothelial cell density (specular microscopy): 1,706 cells/mm² OD, 2,519 cells/mm² OS.

Late Presentation & Findings (April 2026):
Patient presented with a 13-month post-n-DSAEK decline in BCVA OD to 0.2. The graft remained clear, CCT was recorded as 518 μm, and right-eye ECD was 1488 cells/mm².

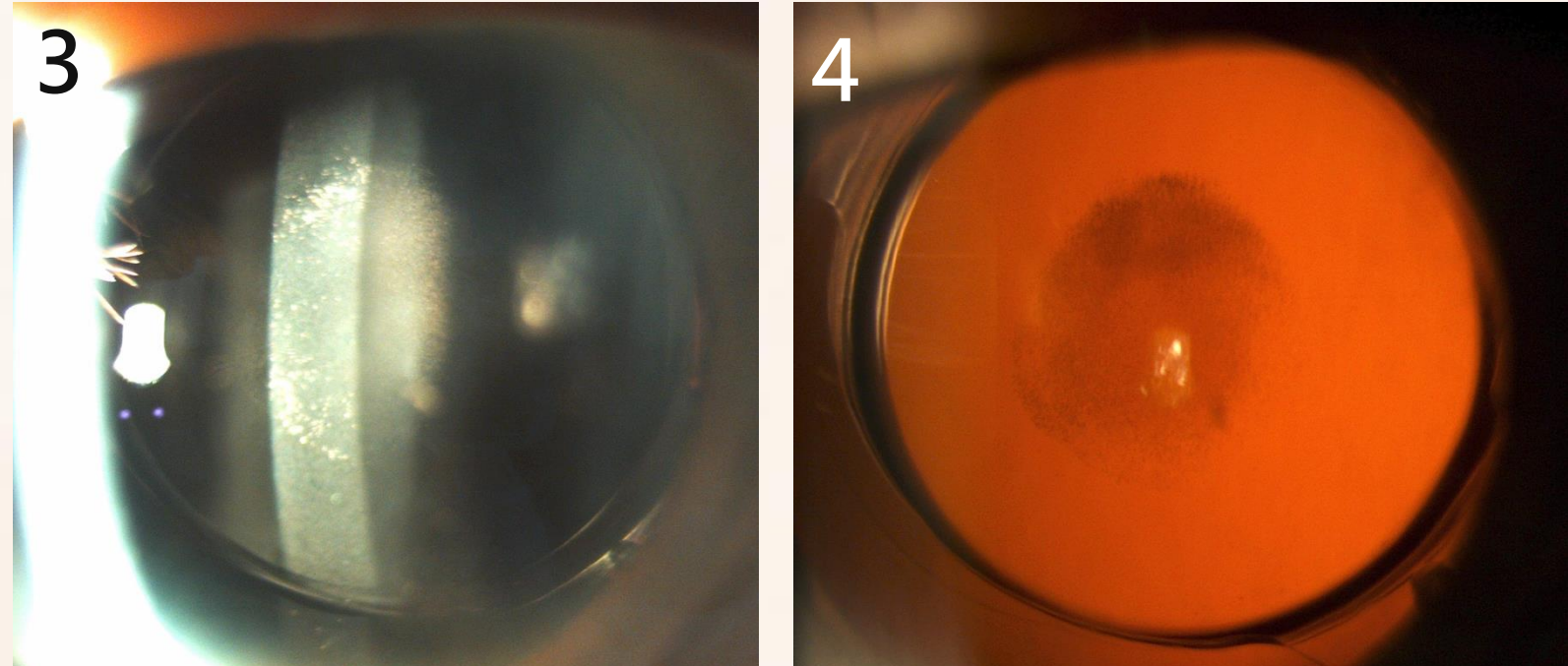
Biomicroscopy: 3-mm central, fine granular opacification on the anterior surface of the hydrophilic acrylic IOL, strictly restricted to the pupillary area.

Anterior-Segment OCT (AS-OCT): Confirmed hyperreflective granular deposits localized precisely to the anterior optic surface.

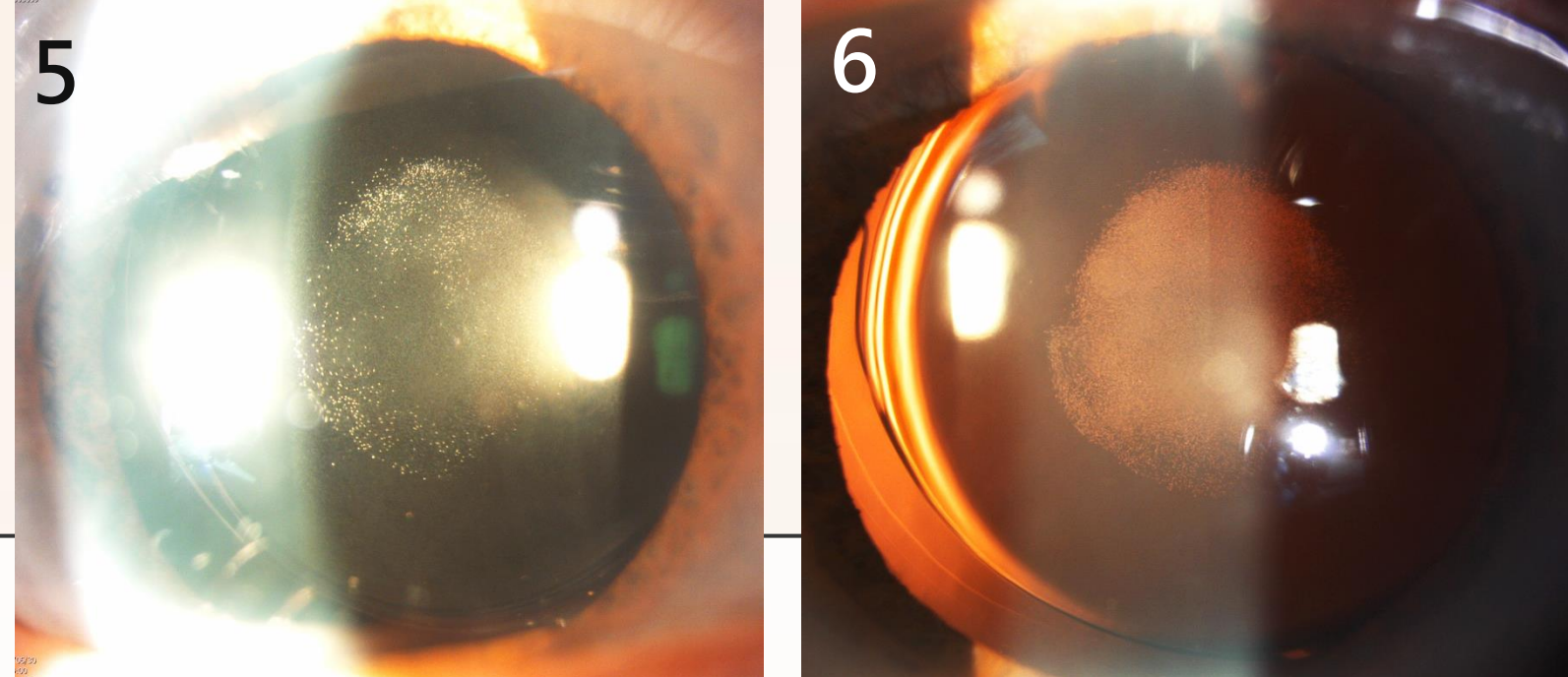
Clinical Course: Opacification and BCVA remained stable at 0.2 through September 2026.



Figures 1–2. February 27, 2025. Preoperative right-eye photographs showing corneal haze and edema on diffuse illumination (Fig. 1) and a clear IOL (Fig. 2).



Figures 3–4. April 7, 2026. Postoperative right-eye slit-lamp and retro-illumination photographs showing central granular IOL opacification.



Figures 5–6. Sep 22, 2026. Right-eye retroillumination and slit-lamp photographs demonstrating stable central granular IOL deposits. BCVA remained 0.2.

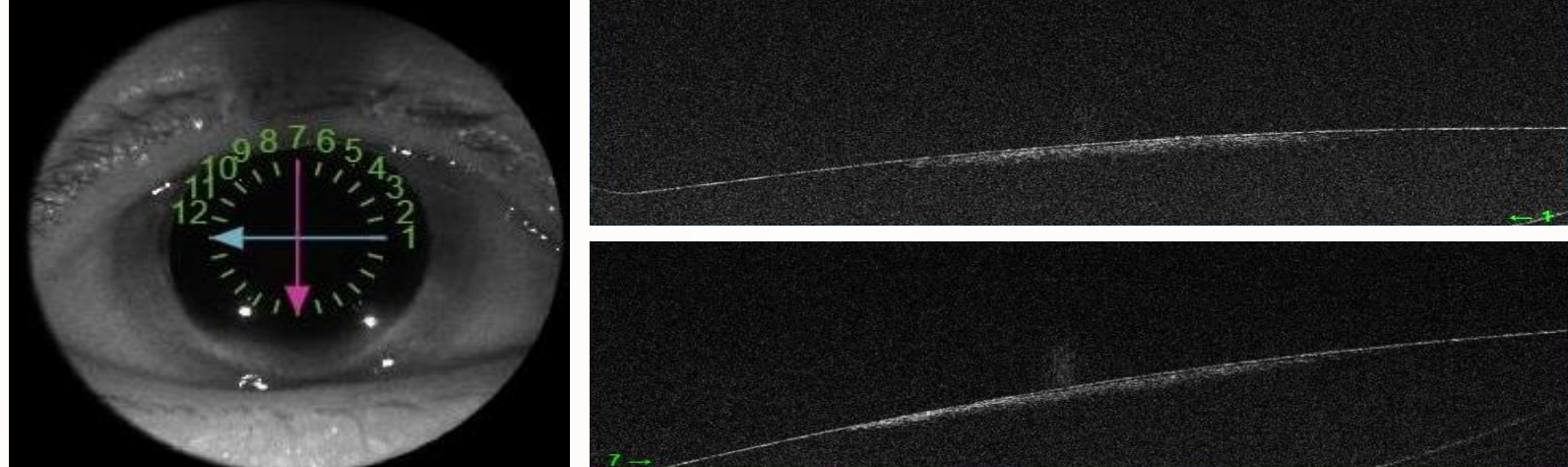


Figure 7. AS-OCT: anterior IOL surface deposits

Discussion

1. Clinical Significance of IOL Opacification

Intraocular lens opacification directly impairs visual acuity and quality of vision. Unlike posterior capsule opacification (PCO), which is routinely managed with neodymium-doped yttrium-aluminum-garnet (Nd:YAG) laser capsulotomy, opacification inherent to the IOL biomaterial cannot be treated with laser therapy. When central calcification causes functional visual loss, surgical IOL explantation and exchange remain the only definitive treatment, carrying inherent surgical risks such as capsular bag tear, vitreous loss, and corneal endothelial cell depletion.

2. Risk Factor Profile

IOL Biomaterial: Hydrophilic acrylic (hydrogel) polymers, owing to their high water content and surface ionic charge interactions, are vastly more susceptible to secondary calcium-phosphate precipitation than hydrophobic acrylic, silicone, or PMMA materials. [1]

Ocular Procedures: Intraocular procedures utilizing gas/air tamponade (n-DSAEK/DSAEK/DMEK, pars plana vitrectomy with gas, or glaucoma filtration surgery) represent the primary surgical triggers. [6]

Systemic & Local Factors: Diabetes mellitus, arterial hypertension, and chronic low-grade inflammation disrupt the blood–aqueous barrier, raising aqueous humor concentrations of calcium, phosphate, and systemic proteins.

3. Proposed Pathogenesis

- a. **Trigger Exposure:** Direct contact between the anterior IOL surface and the intracameral air/gas bubble. [1, 2, 4, 5]
- b. **Surface Barrier Disruption:** Metabolic and mechanical stress alters the hydration state and surface properties of the hydrophilic polymer.
- c. **Nucleation:** Elevated local concentrations of calcium and phosphate ions in the aqueous humor nucleate on or immediately beneath the compromised optic surface. [3]
- d. **Crystal Growth & Clinical Opacification:** Progressive calcium-phosphate deposition forms fine, granular, hyperreflective deposits in the central pupillary zone.

4. Case Correlation & Literature Timeline

In our patient, central anterior optic calcification presented 13 months following n-DSAEK with air injection. This timeline aligns closely with published literature on endothelial keratoplasty with intracameral air/gas, reporting an onset **range of 6 weeks to 43 months** [3]. The pathognomonic central pupillary distribution corresponds directly to the area of **prolonged contact between the buoyant intracameral air bubble and the hydrophilic optic**.

Conclusion

Intracameral air tamponade during endothelial keratoplasty can induce secondary calcification of hydrophilic acrylic IOLs, producing a characteristic fine, granular opacification strictly localized to the central pupillary zone. Slit lamp exam is the primary diagnostic tool, and AS-OCT help confirming hyperreflective material restricted to the anterior surface/subsurface of the IOL optic and ruling out capsular bag pathology. Nd:YAG laser capsulotomy is ineffective and contraindicated. Surgical IOL exchange is reserved for severe visual loss. Most importantly, surgeons should avoid implanting hydrophilic acrylic IOLs in patients with corneal endothelial disease or other conditions prone to future intraocular air/gas procedures.

References

1. Fellman MA, Werner L, Liu ET, et al. Calcification of a hydrophilic acrylic intraocular lens after Descemet-stripping endothelial keratoplasty: case report and laboratory analyses. J Cataract Refract Surg. 2013;39:799–803. <https://pubmed.ncbi.nlm.nih.gov/23541897/>

2. De Cock R, Fajgenbaum MA. Calcification of Rayner hydrophilic acrylic intra-ocular lenses after Descemet’ s stripping automated endothelial keratoplasty. Eye. 2014;28:1383–1384. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4274278/>

3. Darcy K, Apel A, Donaldson M, et al. Calcification of hydrophilic acrylic intraocular lenses following secondary surgical procedures in the anterior and posterior segments. Br J Ophthalmol. 2019;103:1700–1703. <https://pubmed.ncbi.nlm.nih.gov/30798263/>

4. Lee MS, Tsai IL, Tsai CY, et al. Intraocular lens opacification after Descemet’ s stripping automated endothelial keratoplasty. Taiwan J Ophthalmol. 2017;7:160–163. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5637382/>

5. Ma ST, Yang CM, Hou YC. Postoperative intraocular lens opacification. Taiwan J Ophthalmol. 2018;8:49–51. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5890585/>

6. Fernández J, Sánchez-García A, Rodríguez-Vallejo M, Piñero DP. Systematic review of potential causes of intraocular lens opacification. Clin Exp Ophthalmol. 2020;48:89–97. <https://pubmed.ncbi.nlm.nih.gov/31581356/>